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Dr. Michael Rowan
Executive Editor, Royal Society of Chemistry

Subject: Manuscript submission

Dear Dr. Rowan, dear Michael,

We are grateful for your kind invitation to contribute to the *Green Chemistry* 25th Anniversary themed collection. Herein, we submit our paper titled **"Integrating climate policies in the sustainability analysis of green chemicals"** by A. Nabera, A. J. Martín, R. Istrate, J. Pérez-Ramírez, and G. Guillén-Gosálbez for your consideration.

Green routes for chemical production have been consistently reported to be environmentally superior to dominant fossil-based counterparts through life cycle assessments (LCA). However, chemical production routes depend on material and energy inputs from other sectors of the economy. These sectors will be profoundly shaped over the coming decades by the implementation of climate policies emanated from the Paris Agreement. The evolution of the sustainability of green routes in this broader context is still uncertain.

In this work, we tackle this analysis assessing the future environmental performance of three of the main chemicals by volume, namely hydrogen, ammonia, and methanol. They comprise more than 50% of the chemical industry's emissions and are thus representative case studies. In our prospective LCA, we compare fossil and green production routes over the crucial period 2020 to 2050 under three climate policies limiting global warming to either 3.5 °C, 2 °C, or 1.5 °C considering their effects in the power, materials, and transportation sectors. Green routes for hydrogen and ammonia production including wind and solar-based could witness a reduction in emissions of up to 90% by 2050, while methanol production can achieve net CO_2 capture. In contrast, climate policies will have a very limited effect on conventional fossil-based routes, limiting in the best case the impact reduction to 10%. A further analysis could quantify the contributions from other sectors to these trends, revealing, for instance, that the decarbonisation of the power sector will account for approximately 60% of emission reductions, disclosing how deeply green routes will benefit from a more sustainable economy. Taking this into account and considering current marked differences in the energy sector among countries, the study concludes by presenting a strategic worldwide roadmap for the progressive deployment of green routes minimising global emissions. It discloses that South American countries appear as preferred initial locations to commence their implementation in the 2030s decade. From a broader perspective, our study reveals the bright prospects of utilising renewable energy-driven chemical processes, providing unwavering reinforcement for research activities in this field.

A conscientious effort was made to provide clear messages and tools appealing experimental and process systems engineering researchers, aligned with the character of *Green Chemistry* as an interdisciplinary forum. Please find attached the manuscript along with the table of contents entry and the electronic supplementary information.

We look forward to hearing from you.

Best wishes,

Prof. Dr. Gonzalo Guillén-Gosálbez on behalf of all the co-authors.